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Studierichting: Informatica

Oefeningenreeks 6D nr. 20.6

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{n+1}{n!} \\ & \lim_{n \rightarrow +\infty} \left| \frac{\frac{n+2}{(n+1)!}}{\frac{n+1}{n!}} \right| \\ &= \lim_{n \rightarrow +\infty} \left| \frac{n+2}{(n+1)!} \cdot \frac{n!}{n+1} \right| \\ &= \lim_{n \rightarrow +\infty} \left| \frac{n+2}{(n+1)(n+1)} \right| \\ &= \lim_{n \rightarrow +\infty} \frac{n+2}{(n+1)^2} \\ &= 0 \end{aligned}$$

Omdat $0 < 1$, is de reeks convergent volgens het Criterium van d'Alembert.

References